

Experimental Report

Project 1: Fred the Robot Environment

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The following report presents the results of executing the provided agent and driver code on our implementation of the environment for the Fred the Robot game. The collected data was extracted using pandas, and the plots were generated using Matplotlib.

Background: The “Fred the Robot” game

The “Fred the Robot” game is depicted below, where an agent must traverse a 2D grid until it reaches one of two terminal states: state 4 or state 7 [1]. State 4 (the Charging Station) corresponds to the maximum obtainable reward of 25 points, while state 7 (The Pit) corresponds to the minimum obtainable reward of -25 points. All other states correspond to -1 points. A learning agent can improve its performance on this game (measured by total rewards received) by playing the game many times or for multiple episodes. Note that the maximum achievable score per episode is only 21 since Fred will have to travel to at least four non-terminal states before reaching the charging station.

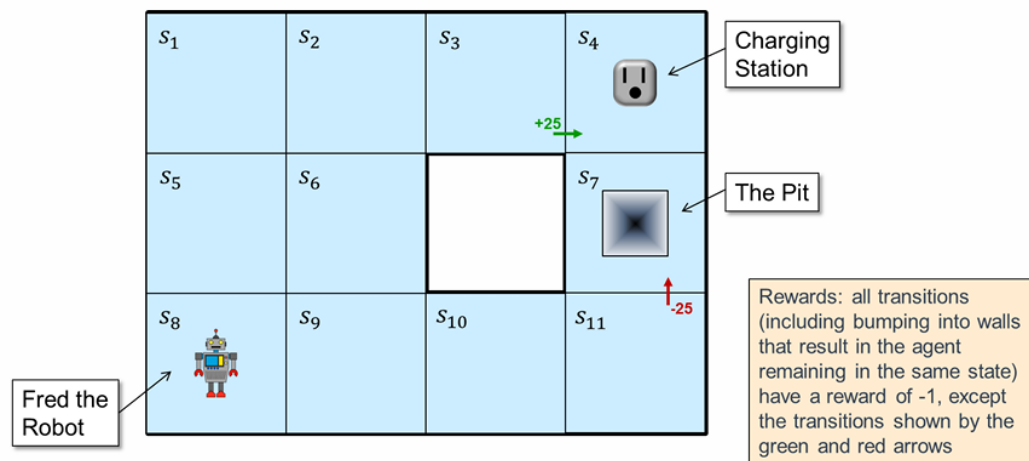
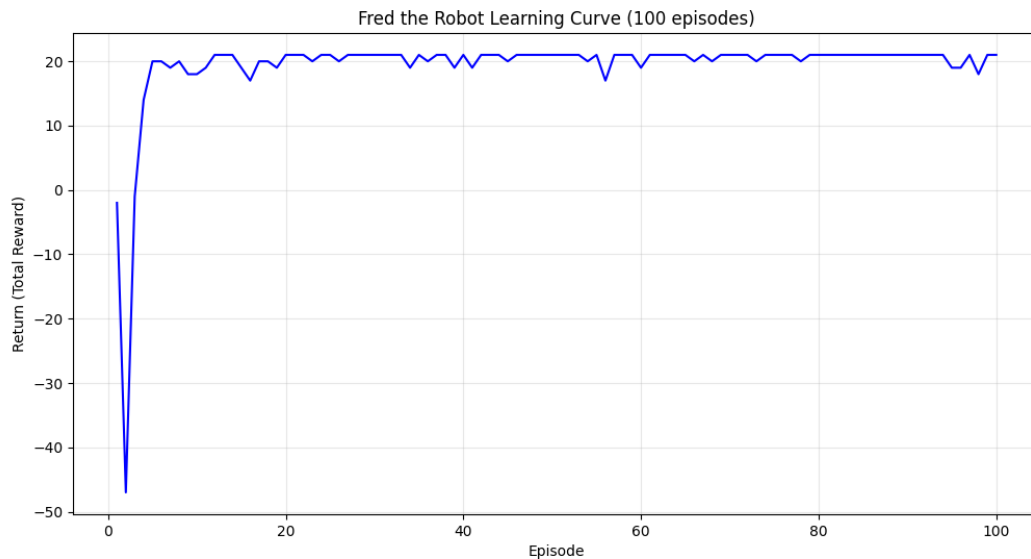
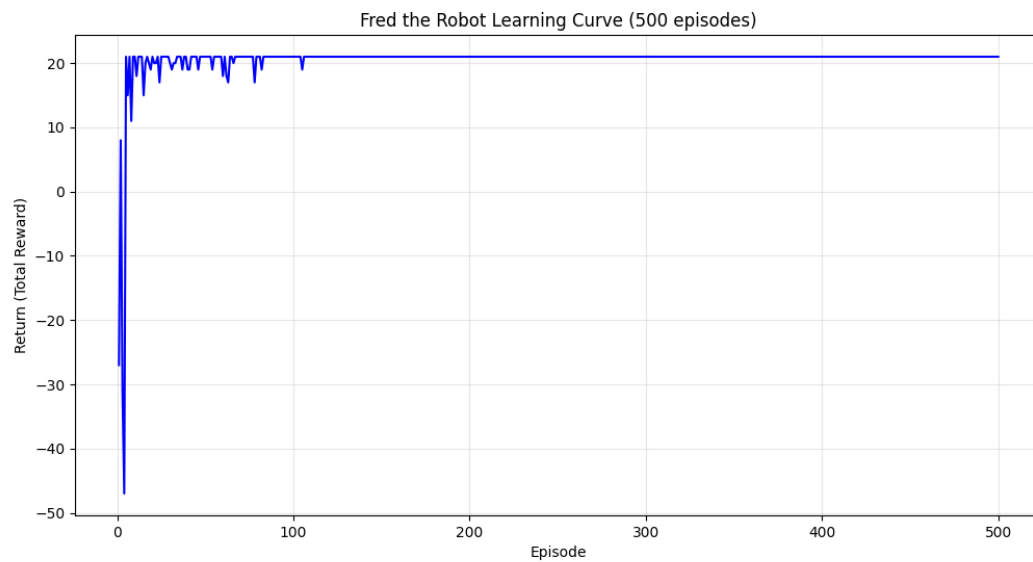


Figure 1: The “Fred the Robot” Puzzle

Experimental Results



Based on the plot above, the provided agent implementation quickly learns to perform well on our simplistic Fred the Robot game. After an initial large drop in performance, you can see the agent's learning quickly recovers and begins to converge after about only 5 episodes. Note the curve appears to converge to 21, which indicates that our agent is correctly learning to maximum its rewards. That said, there is some apparent noise, which corresponds to the stability of our agent's learning. Fortunately, this noise is minimal and can be safely neglected. The plot below shows our agent when trained for 500 episodes. You can see the learning noise disappear after about 110 episodes.



References

[1] Happel, Mark, EN.705.741 Reinforcement Learning, Week 1: Course Introduction, Johns Hopkins University, 2026.